Name _____ Period _____ Date _____

7-2 Lesson Quiz: Evaluating Trigonometric Functions

short form questions keep their Savvas numbers (1, 3, 4); show your work

1. What are the sine and cosine of $\frac{2\pi}{3}$? Write one tile in each box.

$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\sqrt{3}$	4

$\sin\frac{2\pi}{3} =$ ☐ $\cos\frac{2\pi}{3} =$ ☐

Work:

3. Select all the correct statements.

☐ **A.** $\tan\left(-\frac{4\pi}{3}\right) = -\sqrt{3}$

☐ **B.** $\cos\left(-\frac{7\pi}{4}\right) = -\frac{\sqrt{2}}{2}$

☐ **C.** $\cot\left(-\frac{11\pi}{3}\right) = -\frac{\sqrt{3}}{3}$

☐ **D.** $\sec\left(\frac{7\pi}{6}\right) = -\frac{2\sqrt{3}}{3}$

☐ **E.** $\sin\left(\frac{13\pi}{6}\right) = \frac{1}{2}$

Work:

4. Which of the following are true? Select all that apply.

☐ **A.** $\csc 225° = -\sqrt{2}$

☐ **B.** $\cot 225° = \sqrt{2}$

☐ **C.** $\cot 225° = 1$

☐ **D.** $\sec 225° = 1$

☐ **E.** $\sec 225° = -\sqrt{2}$

☐ **F.** $\csc 225° = \sqrt{2}$

Work:

Scoring: Question 1 and Question 4 are 10 points each. Question 3 is a bonus of up to 5 points. Total out of 20.

7-2 Lesson Quiz, short form: answer key

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1. $\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$**,** $\cos\frac{2\pi}{3} = -\frac{1}{2}$**.** Quadrant II, reference angle $\pi - \frac{2\pi}{3} = \frac{\pi}{3}$; terminal point $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.
Distractors: $\frac{1}{2}$ and $-\frac{\sqrt{3}}{2}$ come from using the wrong leg (reference angle $\frac{\pi}{6}$ instead of $\frac{\pi}{3}$, the $\frac{5\pi}{6}$ point); $\sqrt{3}$ is $|\tan\frac{2\pi}{3}|$. Savvas quiz item 1, DOK 1, Skills Review and Practice F84. Taught by Example 4C (the same angle), so this item is a recall check of the packet. **10 points:** 5 per correct box.

3. A, D, E. A: $-\frac{4\pi}{3} + 2\pi = \frac{2\pi}{3}$ (Quadrant II), $\tan\frac{2\pi}{3} = -\sqrt{3}$, true. B: $-\frac{7\pi}{4} + 2\pi = \frac{\pi}{4}$, $\cos\frac{\pi}{4} = +\frac{\sqrt{2}}{2}$, false. C: $-\frac{11\pi}{3} + 4\pi = \frac{\pi}{3}$, $\cot\frac{\pi}{3} = +\frac{\sqrt{3}}{3}$, false. D: $\frac{7\pi}{6}$ is in Quadrant III with reference angle $\frac{\pi}{6}$, $\cos\frac{7\pi}{6} = -\frac{\sqrt{3}}{2}$, $\sec\frac{7\pi}{6} = -\frac{2}{\sqrt{3}} = -\frac{2\sqrt{3}}{3}$, true. E: $\frac{13\pi}{6} - 2\pi = \frac{\pi}{6}$, $\sin\frac{\pi}{6} = \frac{1}{2}$, true. Savvas quiz item 3, DOK 1, F84. Taught by Examples 4 and 5; the coterminal step (adding or subtracting 2π) is Lesson 7-1, and Try It 4b ($-210°$) is the packet's only negative angle. **Bonus 5:** A, D and E and no other box; 3 for two of the three and no wrong box; 0 otherwise.

4. A, C, E. $225°$ is in Quadrant III with reference angle $45°$; terminal point $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$.
$\csc 225° = \frac{1}{-\sqrt{2}/2} = -\frac{2}{\sqrt{2}} = -\sqrt{2}$; $\sec 225° = -\sqrt{2}$; $\cot 225° = \frac{x}{y} = 1$. B, D and F are false (F has the wrong sign). Savvas quiz item 4, DOK 1, F84. Taught by Example 5 with Example 4's Quadrant III signs; the packet Practice ($135°$, TE Practice 14 and Do You Know How 11) is the parallel. **10 points:** all three boxes and no wrong box 10; all three with one wrong box, or two correct and none wrong, 6; one correct and none wrong 3; 0 otherwise.

Items left off this form (available for a retake or Form B): Savvas item 2, "What is $\sin\theta$ if $\cos\theta = \frac{12}{13}$ and θ is in Quadrant IV?" (answer B, $\sin\theta = -\frac{5}{13}$; DOK 1, F84 and F92; the Pythagorean Identity, cut Example 6); Savvas item 5, Melissa's two-way radio, 8-mile radius, Oscar $30°$ west of south (answer $\left(-4, -4\sqrt{3}\right)$: standard-position angle $240°$, $8\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$; DOK 2, F84; any circle centered at the origin, cut Example 7). Item 5's figure (a radius-8 circle on compass axes, grid every 2, a radius $30°$ from the south axis) could be redrawn; it is left off because no kept Example teaches it, not because of the figure.

IXL replacement: SmartScore 70 or above on 6H8 and PAE replaces this quiz with 20/20.

Teacher notes. Every item on the Savvas quiz is DOK 1 except item 5 (DOK 2); the 7-2 TE prints no DOK 3 item anywhere. The TE key for Try It 4 prints $\tan\frac{4\pi}{3} = \frac{\sqrt{3}}{3}$; the correct value is $\sqrt{3}$. The TE key for Practice 14 prints $\sec 135° = \sqrt{2}$; the correct value is $-\sqrt{2}$.